

SoftGrowTech

Python Programming Internship

Internship Report — Task 1

Secure Password Generator Pro

Submitted By

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Internship Organization: SoftGrowTech

Domain: Python Programming

Project Title: Secure Password Generator Pro

Submission Date: July 2026

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1. Introduction

Passwords play an important role in protecting personal and professional digital accounts. Weak passwords can easily be guessed or cracked, making systems vulnerable to unauthorized access.

This project focuses on developing a modern desktop application that generates secure and random passwords using Python. The application allows users to customize password length and character types while providing a simple and user-friendly graphical interface.

2. Project Objective

The objective of this project is to:

- Develop a GUI-based password generator.
- Generate strong and secure passwords.
- Allow users to customize password generation.
- Improve Python programming skills.
- Learn GUI development using CustomTkinter.

3. Project Description

The Secure Password Generator Pro is a desktop application developed using Python and the CustomTkinter library.

The application generates random passwords based on user-selected options such as:

- Password length
- Uppercase letters
- Lowercase letters
- Numbers
- Special symbols

It also includes a password strength indicator, password history, and clipboard copy functionality to improve usability.

4. Features

- Modern graphical user interface
- Adjustable password length
- Uppercase letter option
- Lowercase letter option
- Number option
- Symbol option

- Secure random password generation
- Password strength indicator
- Copy password to clipboard
- Password history
- Clear history feature

5. Technologies Used

Technology	Purpose
Python	Programming Language
CustomTkinter	GUI Development
Random	Password Generation
String	Character Selection
Pyperclip	Copy Password
Tkinter messagebox	User Notifications

6. Project Workflow

The application follows the following workflow:

- User selects password length.
- User selects desired character types.
- User clicks the Generate Password button.
- Application creates a secure random password.
- Password is displayed on the screen.
- Password strength is evaluated.
- Generated password is stored in history.
- User can copy the password or clear the history.

7. Implementation

The project was implemented using Python and CustomTkinter.

The application includes:

- Event-driven programming
- GUI widgets

- Functions
- Conditional statements
- Loops
- Python libraries
- Randomized password generation
- Clipboard integration

The project follows a modular and beginner-friendly coding structure.

8. Screenshots

Screenshot 1 – Main Interface

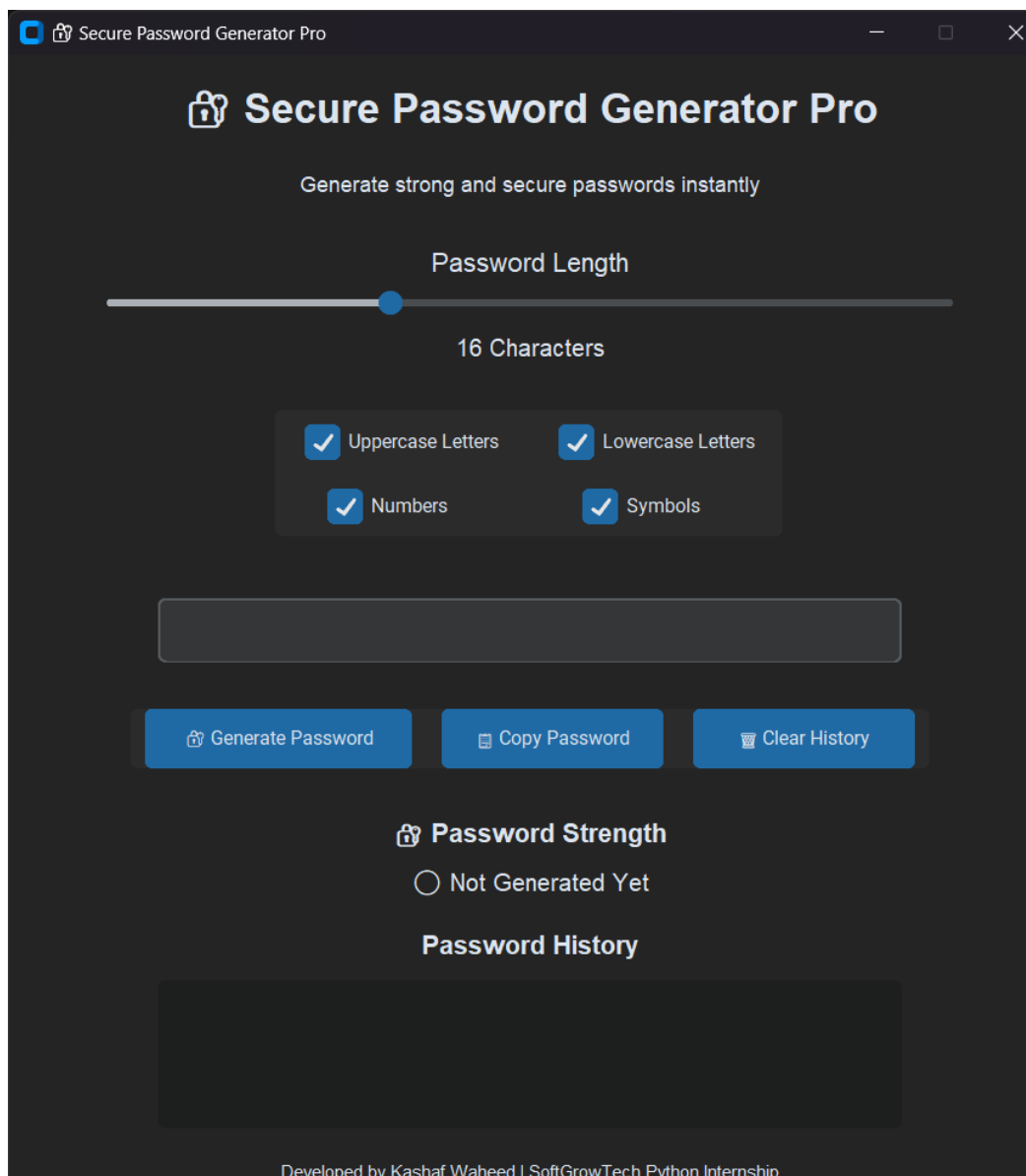


Figure 1: Main interface of Secure Password Generator Pro

Screenshot 2 – Password Generated

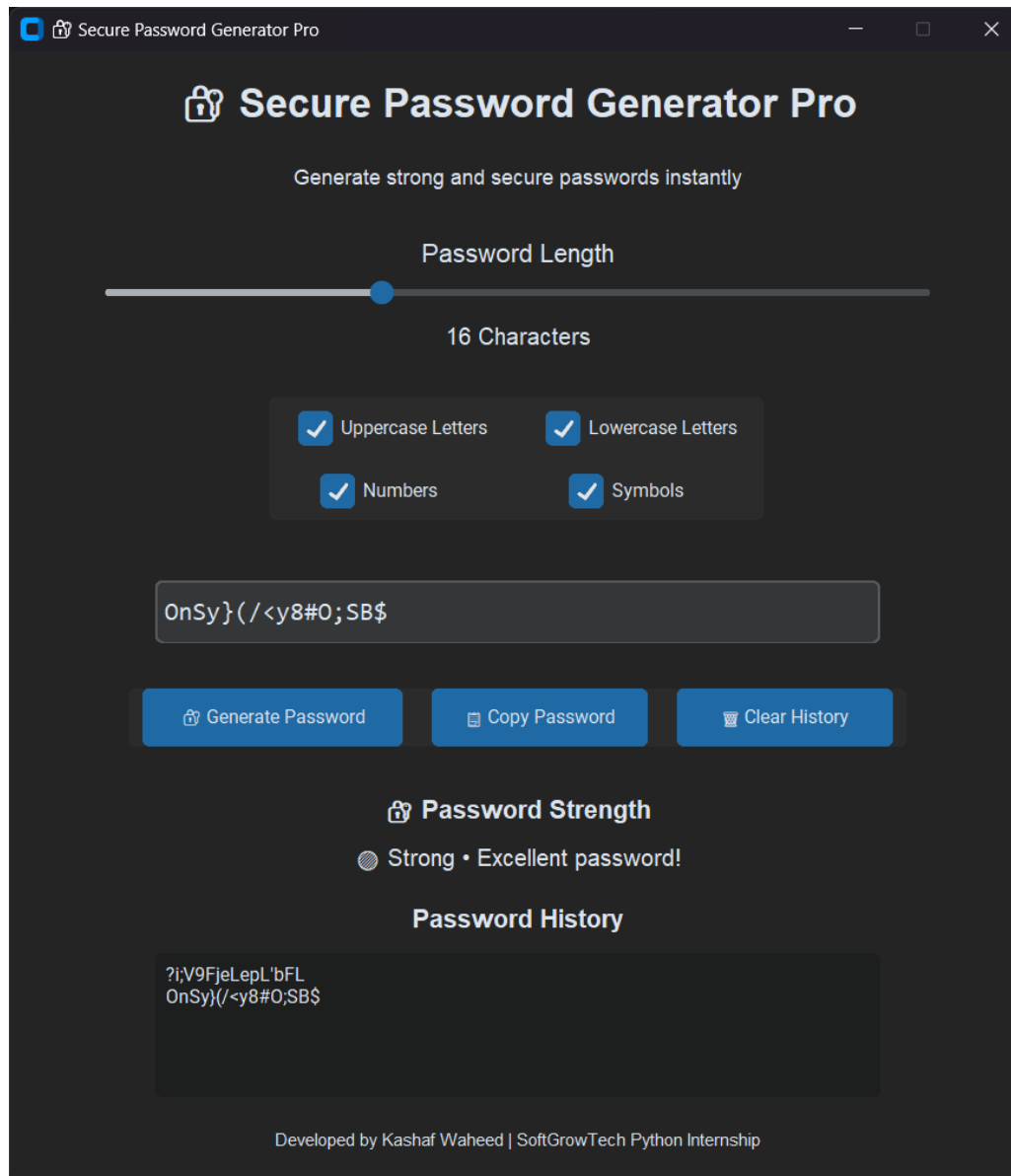


Figure 2: A secure password successfully generated

Screenshot 3 – Password Copied

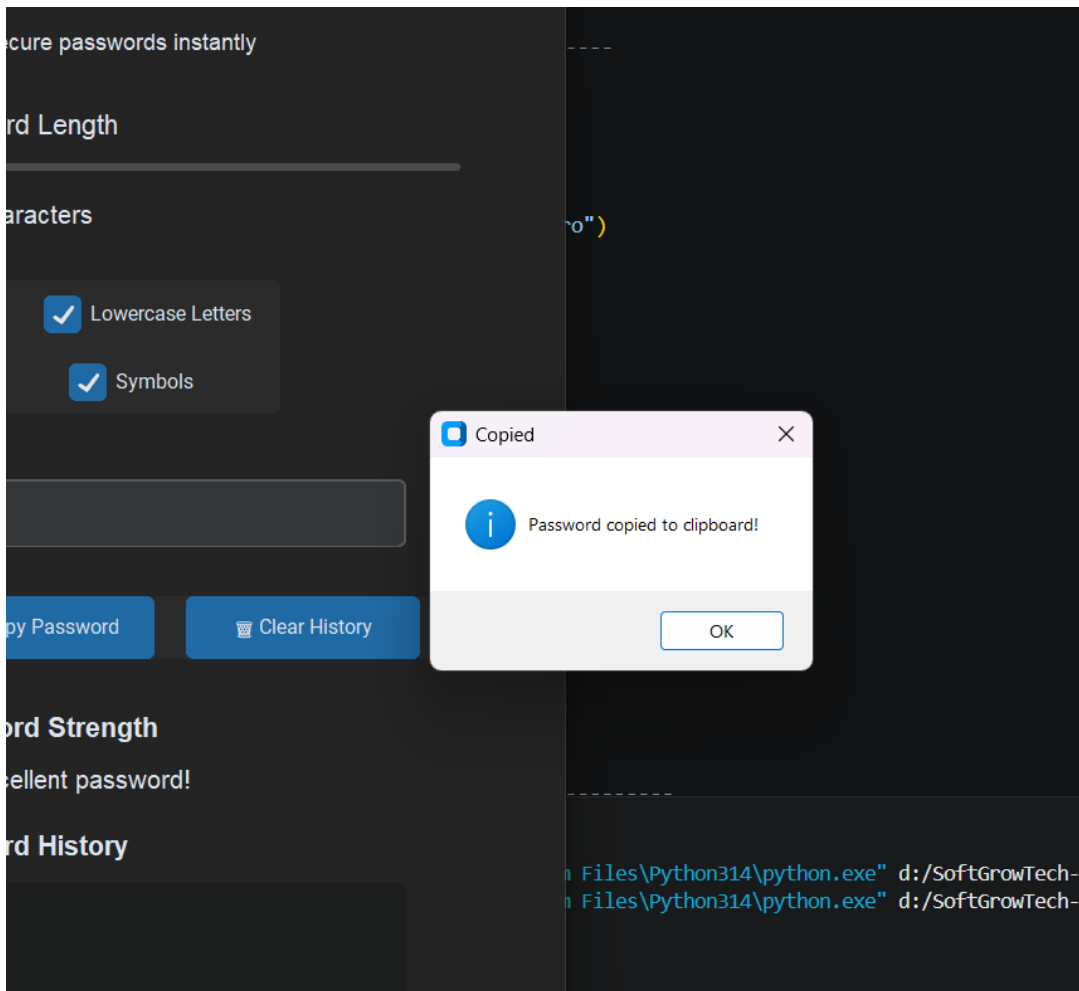


Figure 3: Password copied to clipboard with strength indicator shown

Screenshot 4 – Password History

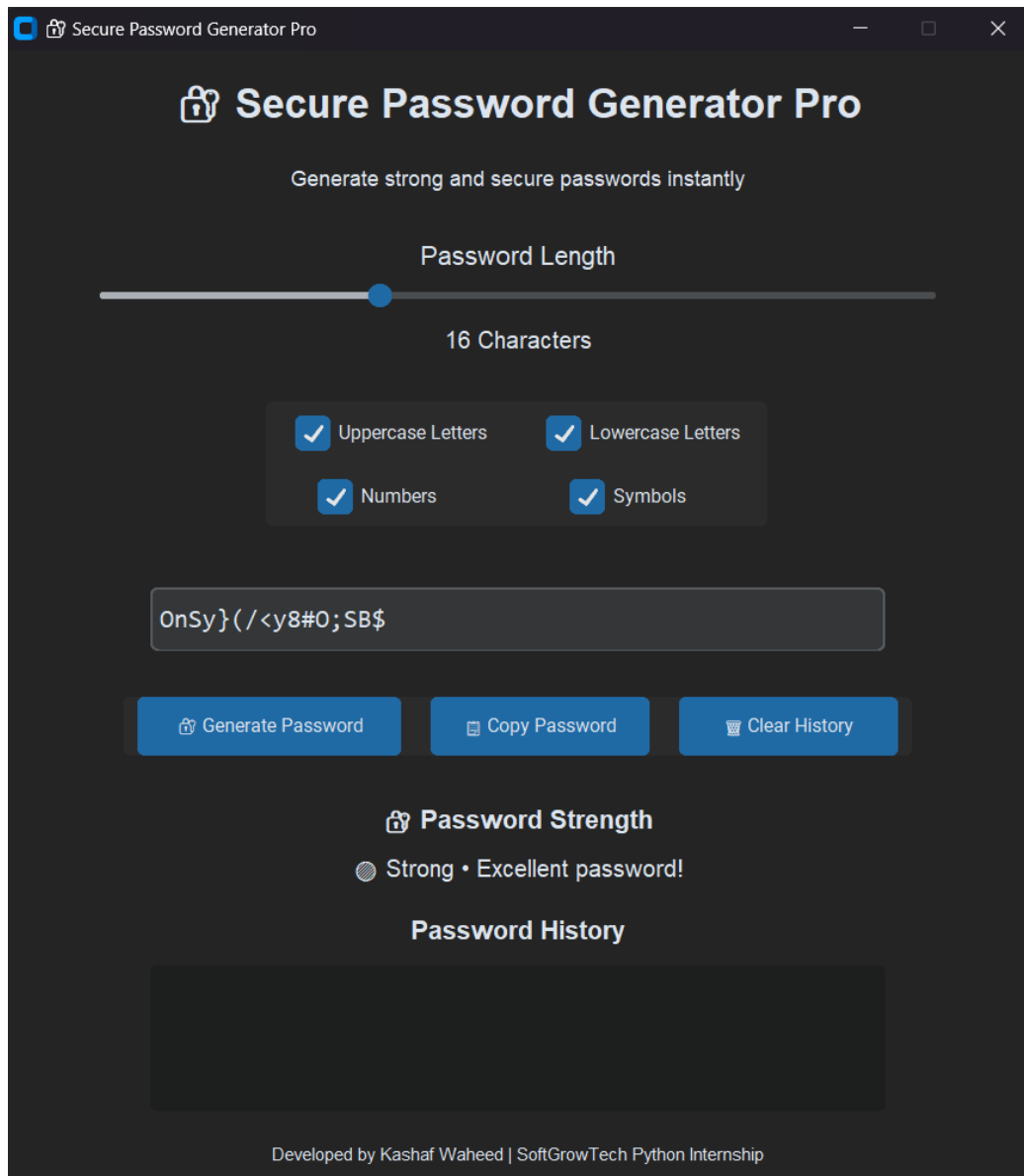


Figure 4: Password history log tracking previously generated passwords

9. Challenges Faced

During the development of this project, several challenges were encountered:

- Learning CustomTkinter widgets.
- Understanding event-driven programming.
- Fixing function definition order issues.
- Implementing the password strength indicator.
- Managing password history.
- Debugging GUI-related errors.

Each challenge improved my debugging skills and understanding of Python GUI development.

10. Learning Outcomes

Through this project, I learned:

- GUI development using CustomTkinter
- Python functions and modules
- Random password generation
- Event handling
- Clipboard operations
- Password security concepts
- Debugging techniques
- Professional project documentation
- GitHub project organization

11. Conclusion

The Secure Password Generator Pro project successfully meets its objectives by providing a modern and user-friendly application for generating secure passwords.

This project enhanced my Python programming skills, introduced me to GUI application development, and strengthened my understanding of password security and software development practices.

The experience gained during this internship has improved my confidence in building practical Python applications and maintaining professional GitHub repositories.

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